

Performance Testing

Date	15 March 2026
Team ID	xxxxxx
Project Name	Rising Water
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter 5.6.3
Type of Testing	Load Testing
Target Module / API	Home Page (GET /)
Test Environment	Hosted Server(Render)
Test Date	30-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home Page Load	10	30sec	Page loads successfully
2	Concurrent Users	25	30sec	Stable response
3	Load Test	50	10sec Ramp-Up	No failures
4	Continuous Requests	50	250	Stable performance

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.2sec	pass	Very fast
2	Response Time (Max)	< 5 seconds	2.738sec	pass	Within limit
3	Throughput (Req/sec)		24.2 req/sec	pass	Good throughput
4	Error Rate	< 1%	0.00%	pass	No errors
5	CPU Utilization	< 80%	N/A	N/A	Not measured by JMeter
6	Memory Utilization	< 80%	N/A	N/A	Not measured by JMeter

Step 4: Observations & Analysis

Key Findings

- The application successfully processed 250 requests without failures.
- The average response time was 0.2 seconds, which is well below the target.
- The maximum response time was 2.738 seconds, remaining within the acceptable limit.
- No request errors were observed during testing.
- The application remained stable throughout the load test.

Bottlenecks Identified

- No significant bottlenecks were observed.
- A few requests took longer than the average under concurrent load, but all remained within acceptable limits.

Optimization Steps Taken

- Deployed the application on Render.
- Configured Gunicorn as the production server.
- Optimized model loading to occur only once at application startup.
- Used Apache JMeter to validate application performance under concurrent requests.

Step 5: Screenshots / Evidence

The following screenshots present the performance testing results obtained using Apache JMeter.

Flood_Test.jmx (C:\Users\GURAVIAH\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Flood_Test.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan
Thread Group
HTTP Request
Summary Report
Aggregate Report
View Results Tree

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: g-jmeter-5.6.3\bin\Flood_Test.jmx Log/Display Only: ☐ Errors ☐ Successes

Label	# Sampl...	Average	Min	Max	Std. Dev.	Error %	Through...	Receive...	Sent KB/...	Avg. Byt...
HTTP Re...	250	200	86	2738	379.44	0.00%	24.2/sec	135.82	3.12	5739.0
TOTAL	250	200	86	2738	379.44	0.00%	24.2/sec	135.82	3.12	5739.0

☐ Include group name in label? ☒ Save Table Header

Flood_Test.jmx (C:\Users\GURAVIAH\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Flood_Test.jmx) - Apache JMeter (5.6.3)

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Test Plan

- Thread Group
 - HTTP Request
 - Summary Report
 - Aggregate Report**
 - View Results Tree

Aggregate Report

Name:

Comments:

Write results to file / Read from file

Filename: Log/Display Only: ☐ Errors ☐ Successes

Label	# Sam...	Avera...	Median	90% Li...	95% Li...	99% Li...	Min	Maxim...	Error %	Throu...	Receiv...	Sent K...
HTTP ...	250	200	104	209	288	2474	86	2738	0.00%	24.2/s...	135.82	3.12
TOTAL	250	200	104	209	288	2474	86	2738	0.00%	24.2/s...	135.82	3.12

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